

DARK MODE PREFERENCE AND ITS IMPACT ON USER COMFORT

A Research Project

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CHAPTER 1

THE PROBLEM AND ITS SETTING

INTRODUCTION

In today's digital age, the use of electronic devices has become an essential part of everyday life. People rely heavily on smartphones, laptops, tablets, and computers for communication, education, work, and entertainment. The rapid advancement of technology has transformed the way individuals interact with information, making digital platforms an integral part of modern society. As digital usage continues to increase, attention has shifted toward improving user experience and visual comfort, particularly in interface design.

One of the most notable developments in interface design is the introduction of dark mode. Dark mode refers to a display setting that presents light-colored text and elements against a dark or black background. This feature has been widely adopted across various operating systems, applications, and websites. Popular platforms such as Facebook, YouTube, Instagram, and operating systems like Windows, Android, and iOS offer dark mode as an option for users. Its growing popularity reflects the demand for personalized and visually comfortable digital experiences.

The increasing amount of time spent on digital devices has raised concerns regarding eye health and visual fatigue. Students, professionals, and even young children spend long hours staring at screens for online classes, work-from-home setups, gaming, and entertainment. This prolonged exposure often leads to symptoms such as dry eyes, headaches, blurred vision, and eye strain. These issues have prompted developers and

researchers to explore alternative display designs that can reduce discomfort and improve overall usability.

Dark mode is commonly believed to reduce eye strain by lowering screen brightness and minimizing glare, especially in low-light environments. Many users claim that it provides a more relaxing viewing experience, particularly during nighttime use. Additionally, dark mode is said to help conserve battery life on devices with OLED or AMOLED screens, making it a practical option for mobile users. Due to these perceived benefits, dark mode has become a preferred display setting for many individuals.

However, despite its popularity, the effectiveness of dark mode remains a topic of debate. Some studies suggest that light mode offers better readability, especially for long texts, as dark text on a light background enhances contrast and legibility. Others argue that dark mode may cause difficulty in reading, especially under bright lighting conditions. These contrasting viewpoints indicate that user preference and environmental factors play a significant role in determining the effectiveness of dark mode.

In the Philippine context, digital device usage has significantly increased, particularly due to the shift toward online learning and remote work. Students now rely heavily on digital platforms for lectures, research, and academic submissions. As a result, many Filipino users have turned to dark mode as a way to reduce visual discomfort during prolonged screen use. Dark mode has also gained popularity due to its modern appearance and perceived health benefits.

Despite its widespread use, there is limited local research that examines the actual impact of dark mode on user comfort and visual performance. Most available studies focus on

foreign populations, which may not fully represent the experiences of Filipino users. Cultural habits, environmental lighting, and device usage patterns differ across regions, making it important to conduct localized studies to better understand user preferences and experiences.

User perception plays a crucial role in determining the effectiveness of dark mode. While some individuals find dark mode relaxing and visually appealing, others experience discomfort and difficulty in reading. These differences highlight the importance of considering individual preferences when evaluating user interface designs. Factors such as age, screen brightness, type of device, and duration of use can significantly influence how users perceive dark mode.

Furthermore, dark mode is not only a functional feature but also an aesthetic choice. Many users associate dark mode with a modern and sleek appearance, which enhances their overall experience. This aesthetic appeal contributes to its widespread adoption, particularly among younger users. However, visual appeal alone does not guarantee usability, making it necessary to examine its impact on comfort and efficiency.

This study aims to analyze the preference for dark mode and its effect on user comfort. It seeks to determine whether dark mode truly helps reduce eye strain and improve user satisfaction or if its popularity is primarily driven by visual trends. By examining user experiences and perceptions, this research aims to provide a clearer understanding of the advantages and limitations of dark mode.

Moreover, this study intends to contribute to the growing field of human-computer interaction by providing insights into how interface design affects user well-being. The

findings of this research may help developers design more user-friendly applications and assist users in making informed decisions about their display preferences.

In conclusion, dark mode has become an essential feature in modern digital interfaces due to its perceived benefits in reducing eye strain and improving visual comfort. However, varying user experiences suggest that its effectiveness depends on several factors, including lighting conditions, user preference, and duration of use. This study seeks to explore these factors in detail and provide evidence-based insights into the impact of dark mode on user comfort. Through this research, the study aims to contribute valuable information that supports the development of healthier and more efficient digital environments.

As digital technology continues to evolve, user expectations regarding interface design also continue to grow. Users are no longer satisfied with systems that are only functional; they now expect interfaces to be visually appealing, comfortable, and adaptable to their personal preferences. This shift in user expectations has encouraged developers to integrate features that promote usability and visual wellness. Dark mode, as one of these features, has become a symbol of modern interface design that prioritizes both aesthetics and comfort.

The increasing awareness of digital eye strain has further contributed to the popularity of dark mode. With the rise of online learning, remote work, and digital entertainment, individuals are exposed to screens for extended periods daily. This prolonged exposure often leads to discomfort, reduced concentration, and fatigue. As a result, users actively seek display settings that can help minimize these effects. Dark mode is commonly

perceived as a solution that lessens eye fatigue by reducing brightness and glare, particularly in dimly lit environments.

Another factor influencing the use of dark mode is its psychological effect on users. Studies suggest that darker interfaces can create a calming and focused atmosphere, which may help users concentrate better on tasks. This is especially beneficial for students and professionals who engage in reading, writing, and data processing for long hours. However, this calming effect may not be experienced by all users, as visual perception varies depending on individual sensitivity, age, and visual acuity.

In educational settings, where students spend a significant amount of time reading digital materials, the choice between dark and light mode becomes especially important. Learning efficiency may be affected by how comfortable students feel while reading on screens. If a display mode causes discomfort or eye strain, it can negatively impact comprehension, focus, and overall academic performance. Therefore, understanding how dark mode affects learners is essential in promoting healthier digital learning environments.

Furthermore, accessibility plays an important role in interface design. Not all users experience visual comfort in the same way. Some individuals, particularly those with visual impairments or astigmatism, may find light text on dark backgrounds difficult to read. This highlights the importance of offering customizable display options rather than enforcing a single mode. Dark mode should be viewed as an alternative rather than a universal solution, allowing users to choose the setting that best suits their needs.

The growing demand for personalization in digital platforms also contributes to the popularity of dark mode. Users enjoy having control over how their interfaces look and feel, and dark mode offers a simple yet effective way to personalize the user experience. This sense of control enhances user satisfaction and engagement, making digital interactions more enjoyable and less tiring.

Despite its advantages, dark mode should not be assumed as a one-size-fits-all solution. Research shows that readability, contrast sensitivity, and environmental lighting significantly influence the effectiveness of display modes. For example, in brightly lit environments, light mode may offer better visibility, while dark mode may be more suitable in darker surroundings. These varying conditions emphasize the need for further research to determine when and how dark mode should be used for optimal comfort.

This study recognizes the importance of examining dark mode not only as a visual preference but also as a factor affecting health, productivity, and user experience. By analyzing user responses and behavior, this research seeks to provide a clearer understanding of how dark mode influences comfort and usability in real-life settings. The findings may help bridge the gap between user perception and actual usability outcomes.

Ultimately, this research aims to contribute meaningful insights into the ongoing discussion about digital interface design. As technology continues to advance, understanding the human aspect of digital interaction becomes increasingly important. By exploring the impact of dark mode on user comfort, this study hopes to support the development of interfaces that are not only visually appealing but also health-conscious and user-centered.

CONCEPTUAL FRAMEWORK

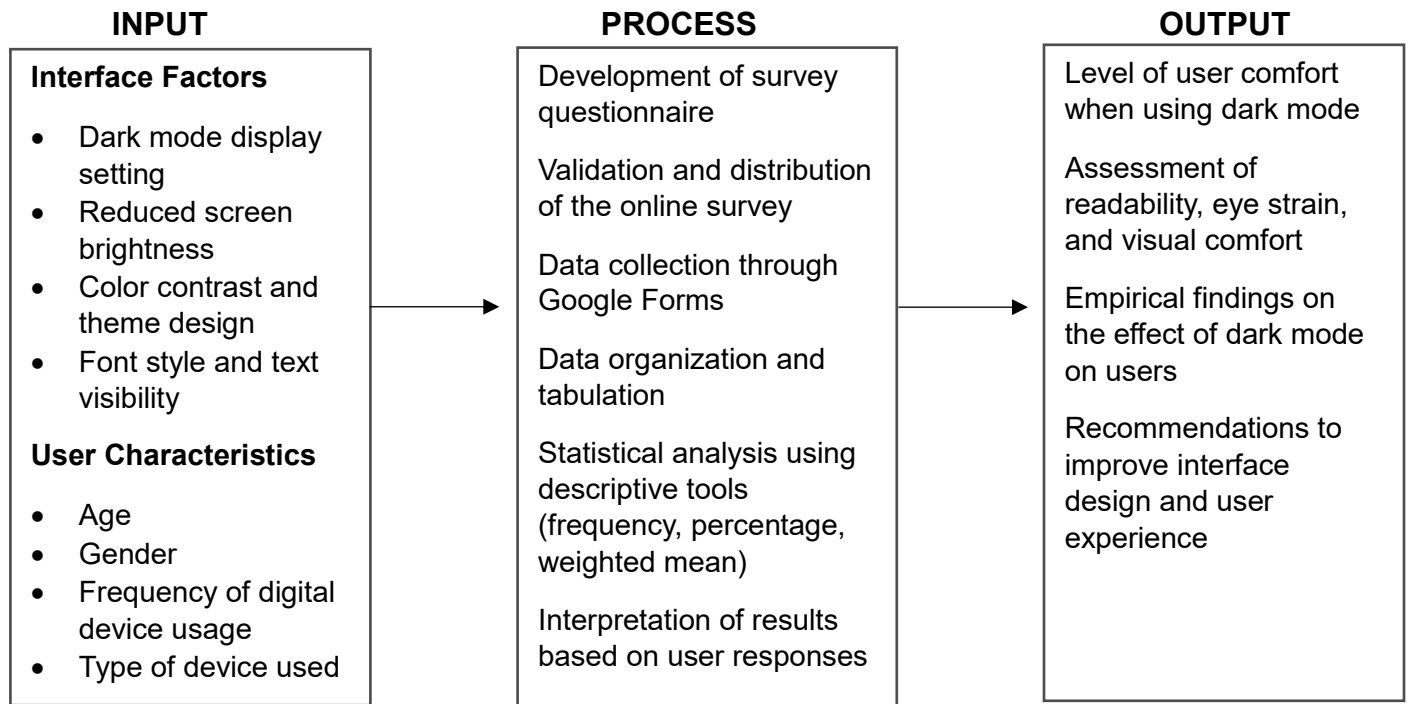


Figure 1. illustrates the conceptual framework of the study, which follows the Input–Process–Output (IPO) Model. This framework presents the logical and systematic flow of the research in determining the effect of dark mode on user comfort. The IPO model is utilized to clearly define the variables of the study, explain how these variables interact through the research process, and identify the expected outcomes based on quantitative analysis.

The Input component of the framework consists of two major categories: interface factors and user characteristics. Interface factors include the dark mode display setting, reduced screen brightness, color contrast and theme design, and font style and text visibility. These elements represent the visual and design-related aspects of dark mode that directly influence how users perceive readability, eye strain, and overall visual comfort.

Proper configuration of these interface factors is essential in reducing visual fatigue, particularly during prolonged screen exposure and usage in low-light environments.

User characteristics are also considered important inputs in the framework, as individual differences may influence the effectiveness of dark mode. These characteristics include age, gender, frequency of digital device usage, and type of device used such as smartphones, laptops, or desktop computers. Differences in screen size, display technology, usage duration, and visual sensitivity may result in varying levels of comfort among users when dark mode is enabled.

The Process component describes the systematic procedures undertaken in the conduct of the study. This process begins with the development of a structured survey questionnaire designed to measure user comfort in terms of readability, eye strain, and visual comfort. The questionnaire is validated to ensure clarity and relevance before being distributed online. Data are collected through Google Forms, allowing efficient and accurate gathering of responses from qualified participants. The collected data are then organized and tabulated, followed by statistical analysis using descriptive tools such as frequency, percentage, and weighted mean. The final stage of the process involves the interpretation of results based on respondents' answers, which serves as the basis for drawing conclusions.

The Output component represents the expected results after the inputs have been processed. These outputs include the level of user comfort when using dark mode, an assessment of readability, eye strain, and visual comfort, empirical findings on the effect of dark mode on users, and recommendations aimed at improving interface design and overall user experience. The outputs are intended to provide evidence-based insights that

may guide users, designers, and developers in making informed decisions regarding the implementation of dark mode.

Overall, the conceptual framework demonstrates how identified inputs are transformed through quantitative research processes into meaningful and evidence-based conclusions regarding the effect of dark mode on user comfort.

THEORETICAL FRAMEWORK

This study is anchored on theories related to visual ergonomics and human–computer interaction, which explain how interface design influences user perception, comfort, and usability. The primary theories supporting this research are the Visual Ergonomics Theory and the Human–Computer Interaction (HCI) Theory.

The Visual Ergonomics Theory emphasizes the importance of designing visual displays that minimize eye strain and promote comfort during prolonged use. According to this theory, factors such as brightness, contrast, color schemes, and text visibility significantly affect visual fatigue and readability. In relation to the study, dark mode is examined as a display feature that may enhance visual comfort by reducing excessive brightness and glare, particularly in low-light conditions where eye strain is more likely to occur.

The Human–Computer Interaction (HCI) Theory focuses on the interaction between users and digital systems, highlighting usability, accessibility, and user satisfaction. This theory supports the inclusion of interface factors and user characteristics in the study, as effective interface design should accommodate diverse users and usage contexts. Evaluating dark mode aligns with HCI principles by examining how users perceive,

interact with, and respond to a specific interface feature in terms of comfort and ease of use.

Together, these theories provide a strong theoretical foundation for the study. They explain why interface factors and user characteristics are essential inputs in evaluating user comfort and justify the expected outcomes related to improved usability and interface design.

Relationship of the Theoretical and Conceptual Frameworks

The theoretical framework provides the foundational concepts that explain how and why dark mode may influence user comfort, while the conceptual framework operationalizes these concepts into measurable variables and research procedures. The theories guide the identification of interface factors and user characteristics, which are translated into survey indicators and analyzed through quantitative methods. Together, the theoretical and conceptual frameworks ensure that the study is both conceptually grounded and methodologically sound.

STATEMENT OF THE PROBLEM

The increasing reliance on digital devices for communication, education, work, and entertainment has resulted in prolonged screen exposure among users. This extended usage has raised concerns regarding eye strain, visual discomfort, and reduced concentration. Dark mode has been introduced as a potential solution to these issues; however, user experiences and perceptions regarding its effectiveness remain inconsistent.

While some users believe that dark mode improves comfort and reduces eye strain, others report difficulty in reading text and experiencing visual discomfort under certain lighting conditions. These conflicting experiences raise questions regarding the actual effect of dark mode on user comfort.

In line with this, the study aims to determine the effect of dark mode on user comfort. Specifically, it seeks to answer the following questions:

1. What is the demographic profile of the respondents in terms of age, gender, and frequency of digital device usage?
2. What is the level of user comfort when using dark mode in terms of readability, eye strain, and visual comfort?
3. Is there a significant effect of dark mode on the comfort level of users?
4. Based on the findings, what recommendations can be proposed to improve user experience when using dark mode?

STATEMENT OF OBJECTIVES

The general objective of this study is to determine the effect of dark mode on user comfort among individuals who regularly use digital devices.

Specifically, the study aims to:

- Determine the demographic profile of the respondents
- Assess the level of user comfort when using dark mode
- Examine the effect of dark mode on user comfort

- Propose recommendations based on the findings

HYPOTHESIS

This research study proposes the following hypotheses:

H1: The use of dark mode has a significant positive effect on users' visual comfort by reducing eye strain and improving readability during prolonged screen usage.

H2: Dark mode reduces the visual and cognitive effort required when viewing digital content, thereby enhancing overall usability.

H3: The display settings and visual elements of dark mode significantly meet user comfort and accessibility needs.

This study assumes that users who find dark mode comfortable and easy on the eyes are more likely to prefer and continue using it.

SCOPE AND LIMITATIONS OF THE STUDY

This study examines the effect of dark mode on user comfort among individuals who use digital devices such as smartphones, laptops, and desktop computers, with focus on readability, eye strain, and visual comfort. The respondents of the study are limited to individuals who have prior experience using dark mode on their digital devices, and the data are collected through an online survey questionnaire distributed using digital platforms.

The study relies on self-reported responses, which may be influenced by personal perception, memory, and honesty of the respondents. The study does not include medical eye examinations, professional vision assessments, or controlled lighting conditions, as

it focuses only on the perceived comfort of users. Additionally, the number of respondents and the sampling method used may limit the generalizability of the findings, and the results may not fully represent the experiences of the entire population. Despite these limitations, the study provides valuable insights that may serve as a reference for future research related to dark mode and visual comfort.

SIGNIFICANCE OF THE STUDY

The findings of this study are significant to various groups as it provides empirical evidence on the effect of dark mode on user comfort, particularly in terms of readability, eye strain, and visual comfort. The results of this research contribute to a better understanding of how display settings influence users' visual experience during prolonged digital device usage.

Users – This study is beneficial to users as it increases awareness of how dark mode affects visual comfort and eye health. The findings may help users make informed decisions when selecting display settings that best suit their visual preferences and usage conditions. By understanding the possible advantages and limitations of dark mode, users may adopt healthier screen habits and reduce the risk of visual discomfort.

Students and Professionals – Students and professionals who rely heavily on digital devices for learning, work, and communication may benefit from the results of this study. The findings provide insights into how dark mode can influence comfort and readability during extended screen exposure, potentially improving focus, productivity, and overall visual well-being.

Developers and Designers – The study serves as a valuable reference for software developers, application designers, and user interface designers. The findings may guide them in designing and optimizing digital interfaces that promote visual comfort, accessibility, and usability. Understanding user responses to dark mode can help improve interface design decisions and enhance user satisfaction.

Educational Institutions – Schools and universities that utilize digital platforms for online learning may use the findings of this study as a basis for recommending appropriate display settings that support students' visual comfort and reduce eye strain during long periods of screen use.

Healthcare and Eye Care Advocates – Although the study does not involve medical diagnosis, the findings may provide useful insights for professionals advocating digital eye health by highlighting user experiences related to eye strain and comfort.

Future Researchers – This study contributes to existing literature on dark mode, visual ergonomics, and human–computer interaction. Future researchers may use the findings as a reference or baseline for further studies, including experimental research designs, comparisons between dark mode and light mode, or investigations involving different populations and devices.

DEFINITION OF TERMS

Dark Mode – A display setting that uses dark-colored backgrounds combined with light-colored text and interface elements to reduce screen brightness, glare, and visual discomfort, especially in low-light environments.

User Comfort – The level of ease, satisfaction, and visual relaxation experienced by users while interacting with a digital interface, particularly during prolonged use.

Eye Strain – A condition characterized by tired, sore, dry, or irritated eyes caused by extended exposure to digital screens, often associated with prolonged focus and improper display settings.

Readability – The ease with which text, symbols, and visual elements can be read, recognized, and understood on a digital display.

Visual Comfort – The degree to which users experience minimal visual fatigue and maintain a relaxed viewing experience when using digital devices.

Interface Factors – The visual and design-related elements of a digital interface, such as brightness, contrast, color scheme, font style, and text visibility, that influence user interaction and comfort.

User Characteristics – The personal attributes of users, including age, gender, frequency of digital device usage, and type of device used, which may affect their perception and experience of dark mode.

Digital Device Usage – The frequency and duration of interaction with digital devices such as smartphones, laptops, and desktop computers for activities including communication, education, work, and entertainment.

Visual Ergonomics – A field of study focused on designing visual displays and work environments that reduce eye strain and promote visual comfort and efficiency.

CHAPTER 2

REVIEW OF LITERATURE AND STUDIES

This chapter presents a review of related literature and studies that are relevant to the present research. It discusses concepts and findings related to dark mode, user comfort, eye strain, readability, and visual comfort. The reviewed literature and studies provide theoretical and empirical support for the variables of the study and help establish the research gap addressed by this paper.

FOREIGN LITERATURE

Dark mode is a user interface design that utilizes dark backgrounds with light-colored text to reduce screen luminance and visual glare. Display polarity plays a significant role in visual performance and comfort, particularly in digital environments. According to Buchner and Baumgartner (2007), negative polarity displays, such as dark mode, can reduce perceived brightness and glare, especially under low ambient lighting conditions, thereby enhancing visual comfort for users. These findings suggest that dark mode may be beneficial in minimizing discomfort during screen use.

Prolonged exposure to digital screens has been associated with visual fatigue and eye strain. Studies have shown that display settings with lower luminance may help reduce visual discomfort during extended reading tasks (Sengsoon & Intaruk, 2025). Their findings support the idea that dark mode can contribute to reduced eye strain, particularly when used on tablet devices and in dim environments.

In addition, user interface design principles emphasize the importance of visual ergonomics in minimizing discomfort during prolonged screen use. Poor contrast,

excessive brightness, and improper color combinations have been linked to increased visual fatigue and reduced user satisfaction. Dark mode addresses these concerns by offering an alternative display setting that lowers overall luminance while maintaining sufficient contrast between text and background. This design approach aligns with ergonomic recommendations aimed at improving visual comfort and reducing eye strain during extended digital interactions.

Furthermore, individual differences in visual perception and environmental lighting conditions play a crucial role in determining the effectiveness of dark mode. While some users benefit from reduced glare and brightness, others may experience difficulty in text recognition depending on font style, contrast level, and screen resolution. These variations highlight the need to evaluate dark mode not only based on technical performance but also on user perception and comfort.

However, readability remains a key concern in dark mode implementation. Research indicates that contrast polarity can influence reading speed and text recognition. Buchner and Baumgartner (2007) noted that while dark mode may improve comfort, light text on dark backgrounds may negatively affect reading efficiency in certain conditions. This suggests that dark mode offers comfort benefits, but its impact on readability may vary depending on individual user characteristics and display settings.

Physiological responses to screen brightness further explain the potential benefits of dark mode. Reduced luminance may lessen pupil constriction, which can contribute to decreased eye fatigue during prolonged screen use (Buchner & Baumgartner, 2007). These findings provide a scientific basis for the increasing adoption of dark mode features in modern digital interfaces.

FOREIGN STUDIES

Several foreign studies have examined user perceptions and ergonomic effects of dark mode compared to light mode interfaces. An experimental study by Pathari (2024) revealed that users generally reported lower levels of eye fatigue and greater comfort when using dark mode on smartphones, particularly in low-light environments. Despite minimal differences in objective performance measures, users consistently expressed a preference for dark mode due to perceived visual comfort.

Similarly, studies on contrast polarity and visual ergonomics indicate that users often favor dark mode because of its perceived ability to reduce glare and eye strain (Sengsoon & Intaruk, 2025). These findings highlight the importance of subjective user perception when evaluating interface design, as comfort and satisfaction may outweigh minor performance differences.

Foreign studies also emphasize that user preference is a significant factor in the acceptance of interface design features. Even when objective performance measures show minimal differences between display modes, users tend to favor interfaces that provide greater comfort and reduced visual fatigue. This preference suggests that perceived comfort may influence sustained usage and overall satisfaction more strongly than minor differences in task performance.

LOCAL LITERATURE

Local literature focusing specifically on dark mode remains limited. However, studies conducted in the Philippines emphasize the increasing use of digital devices among students and professionals. Labana et al. (2020) reported that prolonged screen exposure

among Filipino adolescents is associated with eye strain, headaches, and visual discomfort. Although the study did not focus solely on dark mode, it underscores the importance of screen settings and display features in promoting visual comfort.

With the increasing integration of digital technology in education and professional settings in the Philippines, concerns related to screen-related eye discomfort have become more prominent. Online learning platforms, remote work arrangements, and prolonged use of mobile devices have increased daily screen exposure among Filipino users. These conditions further emphasize the importance of exploring display settings, such as dark mode, that may help alleviate visual strain and enhance user comfort.

Local discussions on digital ergonomics further stress the need for appropriate interface design to reduce visual fatigue and support user well-being. The absence of Philippine-based studies that directly examine dark mode usage highlights a gap in local research that warrants further investigation.

LOCAL STUDIES

Most local studies in the Philippines primarily focus on general digital eye strain and the effects of prolonged screen exposure rather than comparing specific interface modes such as dark mode and light mode. Existing research indicates that Filipino users increasingly rely on digital devices for academic, professional, and social activities, resulting in extended screen time and heightened visual demands (Labana et al., 2020). This growing reliance on digital technology reinforces the need to explore display features that may help alleviate visual discomfort among users.

Despite the growing relevance of digital ergonomics, limited empirical studies in the local context have examined the role of specific interface features in reducing visual discomfort. Most existing studies focus on general health effects of screen exposure rather than evaluating user perceptions of display modes. This limitation highlights the need for localized research that investigates how interface design choices, such as dark mode, influence user comfort and visual experience.

SYNTHESIS OF THE REVIEWED LITERATURE AND STUDIES

The reviewed foreign and local literature suggests that dark mode has the potential to enhance user comfort by reducing screen brightness and glare, particularly in low-light environments. While foreign studies provide substantial evidence supporting the comfort-related benefits of dark mode, findings regarding readability and performance remain inconsistent. Local studies emphasize the prevalence of digital eye strain among Filipino users but reveal a lack of focused research on dark mode usage. This research gap supports the need for the present study, which aims to evaluate user comfort when using dark mode among Filipino digital device users.

Considering both foreign and local findings, it is evident that visual comfort is a multifaceted concept influenced by display settings, environmental conditions, and individual user preferences. While foreign studies provide strong empirical support for the ergonomic benefits of dark mode, local literature underscores the increasing prevalence of screen-related discomfort without offering specific solutions. Bridging this gap requires studies that combine ergonomic principles with user-centered evaluation.

CHAPTER 3

METHODOLOGY

RESEARCH DESIGN

This study employs a quantitative descriptive research design. This design is appropriate because the study aims to gather numerical data and describe the perceptions of users regarding their level of comfort when using dark mode. The quantitative approach allows the researcher to analyze responses objectively using statistical tools, while the descriptive method focuses on describing the existing conditions without manipulating any variables. This research design is suitable for studies that examine user perceptions, behaviors, and experiences through survey questionnaires.

SOURCE OF DATA

The data for this study were obtained from individuals who regularly use digital devices and have prior experience using dark mode features. These respondents were considered appropriate sources of data as they provided relevant and meaningful information necessary to address the research objectives, particularly in understanding user comfort in terms of readability, eye strain, and overall visual comfort. The selection of respondents was suitable for gathering insights based on users' actual experiences with dark mode in their daily digital device usage.

The study employed convenience sampling, a non-probability sampling technique commonly used in quantitative descriptive research. This method allowed the researcher to efficiently collect data from readily available respondents through an online platform,

making it appropriate for survey-based studies that focus on user perceptions and experiences.

Primary Source. The primary data for this study were obtained from the responses gathered through an online survey questionnaire administered via Google Forms. These data included respondents' demographic information such as age, sex, primary device used, and average daily screen time, as well as their responses to statements related to their level of comfort when using dark mode. The primary data reflected the respondents' direct perceptions and experiences and served as the main basis for data analysis and interpretation.

Secondary Source. The secondary data for this study consisted of related literature, existing academic research, journal articles, and credible online publications that discussed dark mode, digital eye strain, visual ergonomics, and user interface design. These sources provided theoretical foundations and supporting evidence that guided the development of the research instrument and helped contextualize and support the findings of the study.

RESEARCH INSTRUMENT

The study utilized quantitative research techniques to obtain relevant data necessary for determining users' perceptions of dark mode and its effect on user comfort. The primary research instrument used was an online quantitative survey questionnaire administered through Google Forms. This instrument was designed to systematically collect measurable data related to readability, eye strain, visual comfort, and overall user experience when using dark mode on digital devices.

Online Survey Research. Online survey research involves the use of internet-based platforms to collect numerical data through structured questionnaires. This method serves as a practical alternative to traditional paper-based surveys due to its accessibility, efficiency, and ability to reach a wider range of respondents across different locations. Online surveys allow for real-time data collection and automatic recording of responses, reducing time and resource constraints while ensuring accuracy in data gathering. This approach is widely used in quantitative studies that aim to analyze trends, patterns, and perceptions using statistical methods.

Quantitative Survey Questionnaire. The online quantitative survey questionnaire was composed of two main sections. The first section gathered demographic information from the respondents, including age, sex, primary device used, and average daily screen time. The second section consisted of ten (10) structured statements related to dark mode usage and user comfort, focusing on readability, eye strain, visual comfort, focus, glare reduction, comfort in low-light environments, and overall satisfaction. A five-point Likert scale was used to measure the respondents' level of agreement with each statement, where 5 represented Strongly Agree, 4 Agree, 3 Neutral, 2 Disagree, and 1 Strongly Disagree. This scale enabled the responses to be quantified and analyzed using appropriate statistical techniques.

DATA GATHERING PROCEDURE

The data gathering process was conducted entirely online. The researcher prepared and finalized the survey questionnaire and embedded an informed consent statement at the beginning of the Google Forms survey. The consent form explained the purpose of the study, the voluntary nature of participation, and the respondents' rights.

After finalization, the survey link was distributed through online platforms such as social media and messaging applications. Respondents were given sufficient time to complete the questionnaire at their convenience. All responses were automatically recorded and compiled by Google Forms. After the data collection period, the responses were reviewed, organized, and prepared for analysis.

DATA ANALYSIS TOOL

The data collected from the online survey were analyzed using appropriate descriptive statistical tools to ensure accurate and systematic interpretation of the results. Descriptive statistics were employed to summarize, organize, and present the data in a manner that is easy to understand and relevant to the objectives of the study. Frequency counts and percentage distribution were used to describe the demographic profile of the respondents in terms of age, sex, primary device used, and average daily screen time. These tools provided a clear overview of the characteristics of the respondents and helped identify patterns in device usage and screen exposure.

STATISTICAL TREATMENT OF DATA

This section presents the statistical tools and procedures used in analyzing the data gathered from the respondents. The collected data were systematically organized, tabulated, and analyzed to address the objectives of the study and to determine the respondents' demographic profile and perceptions regarding the use of dark mode on digital devices. The statistical treatment of data ensured that the results were presented in an objective, accurate, and meaningful manner.

The analysis focused on both descriptive and perceptual data. Descriptive statistics were used to summarize respondent characteristics, while the weighted mean was applied to evaluate respondents' level of agreement with statements related to visual comfort, eye strain, readability, focus, and overall satisfaction when using dark mode. The use of these statistical tools made it possible to identify trends, compare responses, and draw valid conclusions based on the gathered data.

Frequency and Percentage Distribution

Frequency counts and percentage distribution were used to describe the profile of the respondents in terms of age, sex, primary device used, and average daily screen time. These statistical measures helped present the distribution of respondents across different categories and provided a clear picture of the sample population. Frequency counts showed the number of respondents under each category, while percentage distribution allowed for easier comparison of proportions among groups.

The results obtained from frequency and percentage distribution were presented using column charts to enhance clarity and visual understanding. Graphical presentation supported the textual discussion by highlighting patterns and differences among respondents. These tools were essential in analyzing the demographic characteristics of the respondents, which served as a foundation for interpreting their perceptions of dark mode usage discussed in Chapter 4.

Weighted Mean

The weighted mean was used to determine the overall level of agreement of the respondents for each statement related to dark mode usage. This statistical tool was

applied to summarize the respondents' perceptions regarding visual comfort, eye strain, readability, focus, and overall satisfaction.

The weighted mean was computed using the formula:

$$\text{Weighted Mean (WM)} = \frac{\sum(f \times w)}{N}$$

Where:

- WM = Weighted Mean
- f = Frequency of responses
- w = Weight assigned to each response
- N = Total number of respondents

Likert Scale Used

The study utilized a five-point Likert scale to measure the respondents' level of agreement with each statement:

Scale	Description
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

Interpretation of Weighted Mean

To interpret the computed weighted mean values, the following scale was used:

Weighted Mean Range	Description
4.21 – 5.00	Strongly Agree
3.41 – 4.20	Agree
2.61 – 3.40	Neutral
1.81 – 2.60	Disagree
1.00 – 1.80	Strongly Disagree

Use of Graphical Presentation

Graphical presentation was used in this study to enhance the clarity and understanding of the survey results. Column charts were utilized to visually present the data for both the respondents' demographic profile and their perceptions toward dark mode usage. The use of column charts allowed for easier comparison among different response categories and made patterns and trends more visible to the reader.

Through graphical presentation, complex numerical data were transformed into visual representations that are easier to interpret. This approach helped highlight differences in responses across age groups, device usage, screen time, and levels of agreement with each perception statement. The charts also served as visual support for the discussion, analysis, and interpretation of data presented in Chapter 4, enabling readers to better comprehend the findings of the study.

Explanation for the Use of Statistical Tools

The use of frequency counts, percentage distribution, and weighted mean allowed the researcher to systematically analyze both descriptive and perceptual data gathered from the respondents. Frequency counts and percentage distribution were used to describe the demographic characteristics of the respondents and to determine how responses were distributed across different categories.

The weighted mean was employed to summarize the respondents' level of agreement for each survey statement related to dark mode usage. This statistical tool provided a clear numerical value that represented the overall tendency of responses and allowed for meaningful verbal interpretation. By using these statistical tools, the researcher was able to identify patterns, trends, and general perceptions among respondents. The results served as a reliable basis for the discussion of findings, formulation of conclusions, and development of recommendations presented in the succeeding chapters of the study.

ETHICAL CONSIDERATIONS

This study adhered to ethical research standards and complied with the Data Privacy Act of 2012 (Republic Act No. 10173) of the Philippines. An informed consent statement was provided at the beginning of the online survey to inform respondents about the purpose of the study, the voluntary nature of participation, and their right to withdraw at any time without penalty. Personal information collected, such as age and sex, was used solely for academic purposes. All data were treated with confidentiality, securely stored, and accessed only by the researcher. By proceeding with the survey, respondents voluntarily agreed to participate under the stated conditions.

CHAPTER 4

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents, analyzes, and interprets the data gathered from the survey conducted for the study. The presentation of results is aligned with the research objectives and emphasizing clarity, systematic discussion, and interpretative depth. The data are presented using tables and figures, followed by detailed explanations and interpretations to describe respondents' perceptions and experiences regarding dark mode usage.

The analysis focuses on three major areas: (1) demographic profile of the respondents, (2) device usage and screen exposure, and (3) respondents' level of agreement on statements related to dark mode in terms of preference, eye strain, readability, visual comfort, focus, and overall satisfaction.

DEMOGRAPHIC PROFILE OF THE RESPONDENTS

Age Distribution of Respondents

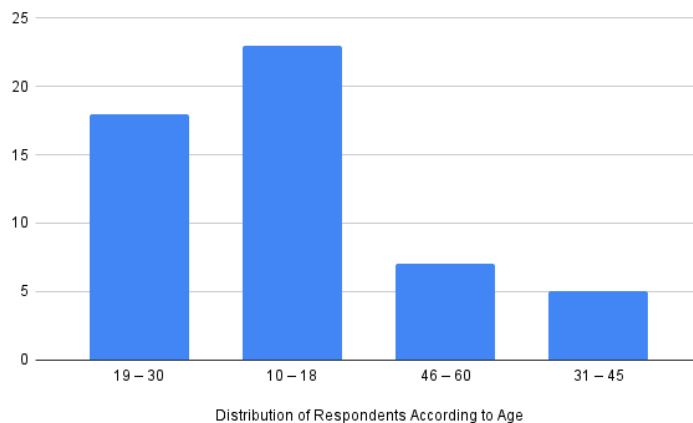


Figure 1.1 illustrates that the respondents are largely concentrated in the younger age groups. This suggests that the study primarily reflects the perceptions of individuals who are

more exposed to digital technology and frequent device usage. Younger users are generally more adaptable to interface features such as dark mode and are more likely to

experience prolonged screen exposure, making their responses particularly relevant to the study.

Gender Distribution of Respondents

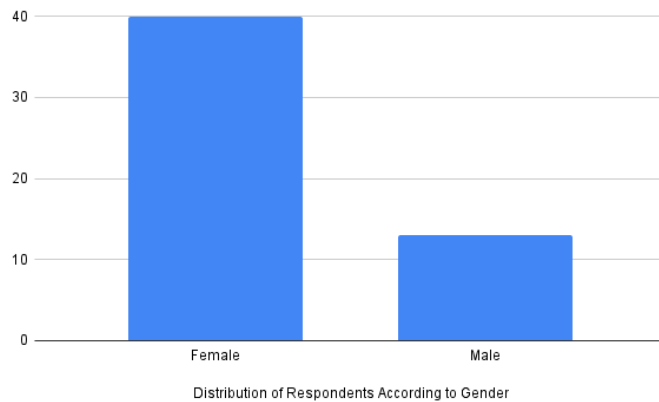


Figure 1.2 shows a higher representation of female respondents compared to male respondents. This indicates that the overall findings may reflect stronger female perspectives regarding visual comfort and usability.

Differences in visual sensitivity and aesthetic preference may influence how display modes are perceived by users.

DEVICE USAGE AND SCREEN EXPOSURE

Primary Device Used

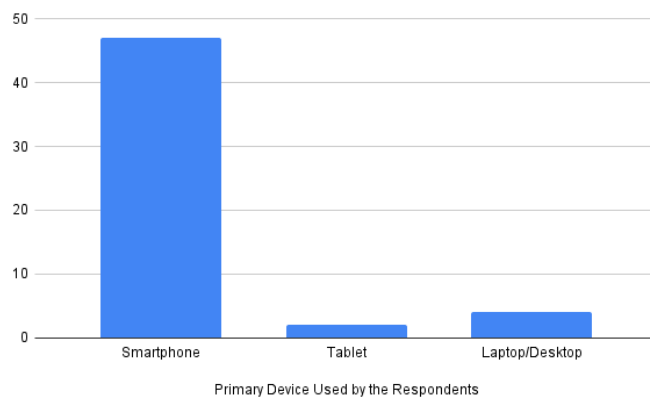


Figure 2.1 reveals that smartphones are the most commonly used devices among respondents. This suggests that dark mode plays a significant role in mobile device usage, where screens are frequently used for long periods

and under varying lighting conditions. The result emphasizes the importance of optimizing dark mode for mobile interfaces.

Average Daily Screen Time

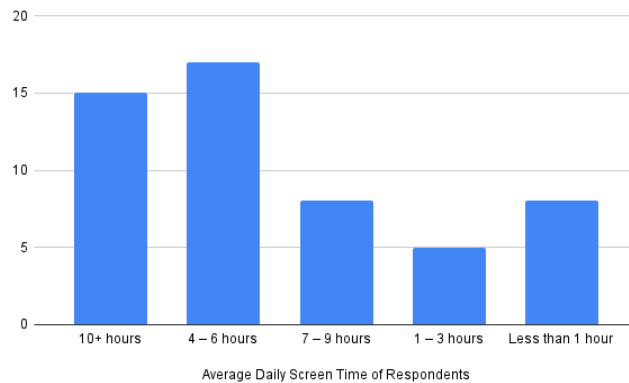


Figure 2.2 indicates that many respondents spend extended hours using digital devices each day. Prolonged screen exposure increases the likelihood of eye strain and visual fatigue, highlighting the relevance of

examining dark mode as a feature that may enhance visual comfort.

LEVEL OF AGREEMENT ON DARK MODE USAGE

The following sections present the respondents' level of agreement on statements related to dark mode. Each figure is discussed individually to provide a clear and comprehensive interpretation of the results.

Preference for Dark Mode Over Light Mode

1. I prefer using dark mode over light mode on my device.

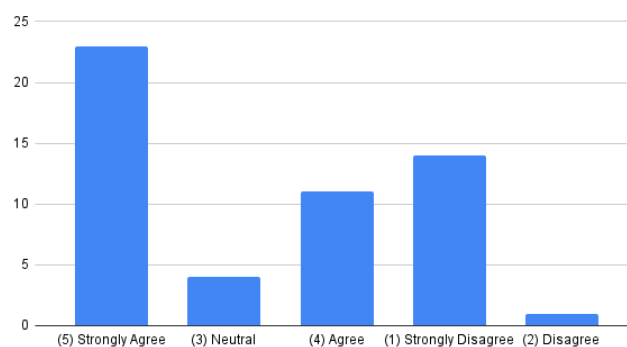
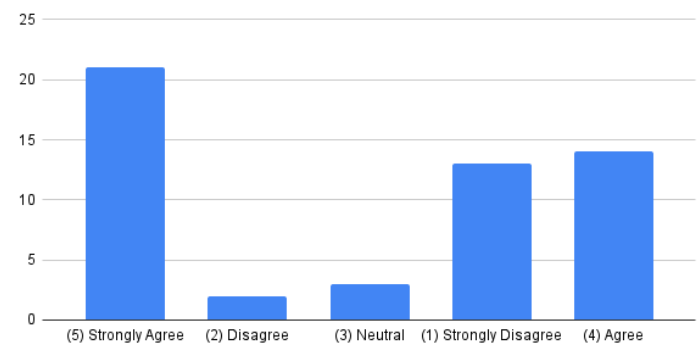


Figure 3.1 suggests that dark mode is generally preferred by a large portion of respondents. However, the presence of users who favor light mode implies that display preferences vary depending on individual comfort and usage conditions.

This highlights the importance of allowing users to choose between display modes.

Dark Mode and Eye Strain Reduction

2. Dark mode helps reduce eye strain when I use my device for long periods.

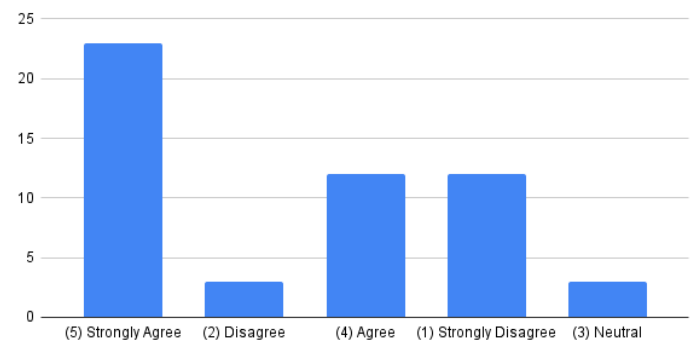


improved visual comfort.

Figure 3.2 indicates that respondents generally perceive dark mode as effective in reducing eye strain during prolonged device usage. This suggests that reduced brightness and glare contribute to

Visual Comfort When Using Dark Mode at Night

3. I feel more visually comfortable when using dark mode, especially at night.



in dark surroundings.

Figure 3.3 shows that respondents feel more visually comfortable when using dark mode in nighttime or low-light environments. This implies that dark mode effectively minimizes discomfort caused by bright screens

Readability of Text in Dark Mode

4. Text in dark mode is easy for me to read and understand.

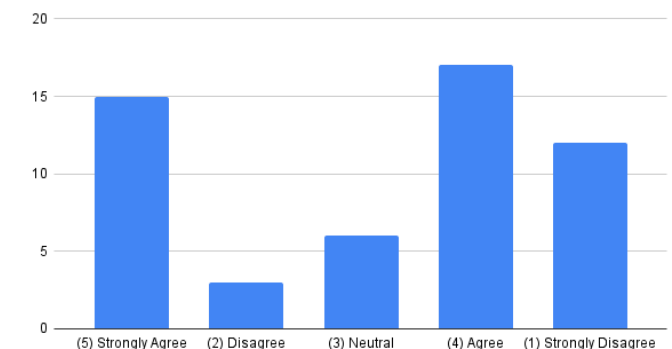


Figure 3.4 reveals mixed perceptions regarding text readability in dark mode. While many respondents find text easy to read, others experience

difficulty, suggesting that readability is influenced by contrast, font design, and individual visual sensitivity.

Dark Mode and User Focus

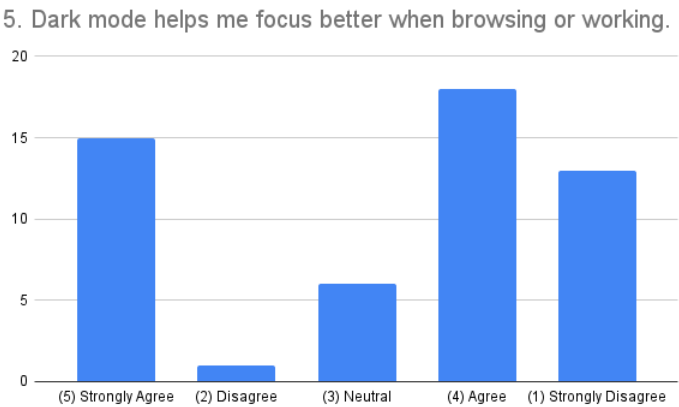


Figure 3.5 suggests that dark mode helps improve user focus during browsing or work-related tasks. Reduced visual distractions and lower brightness may contribute to better concentration and productivity.

Screen Glare in Dark Mode

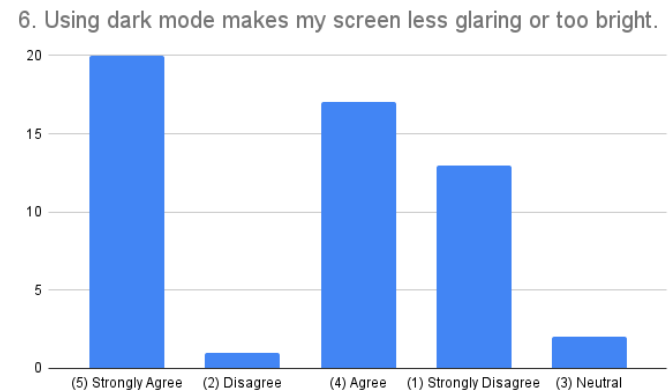


Figure 3.6 indicates that dark mode reduces screen glare. Lower glare levels enhance visual comfort, particularly for users who engage in prolonged screen use.

Eye Relaxation when Using Dark Mode

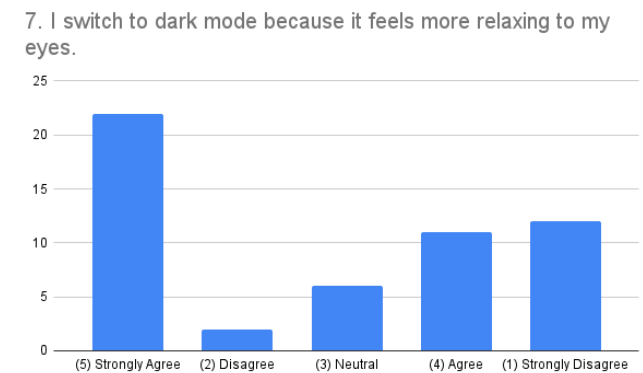


Figure 3.7 suggests that respondents associate dark mode with a more relaxing visual experience. This implies that dark mode may help reduce eye fatigue during extended device usage.

Discomfort of Light Mode in Low-Light Environments

8. Light mode becomes uncomfortable or too bright in low-light environments.

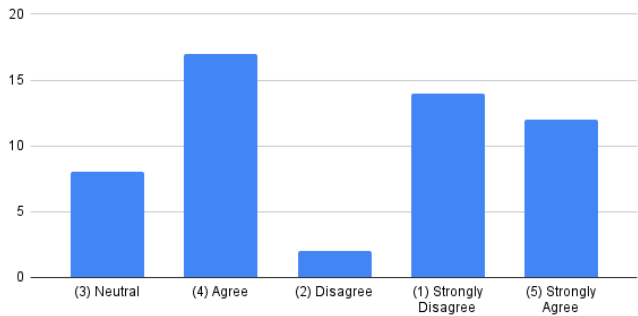


Figure 3.8 indicates that light mode is often perceived as uncomfortable in low-light settings. Bright backgrounds in dark environments may cause visual discomfort, reinforcing the preference for dark mode under such conditions.

Dark Mode and Overall User Experience

9. Dark mode improves my overall experience when using digital devices.

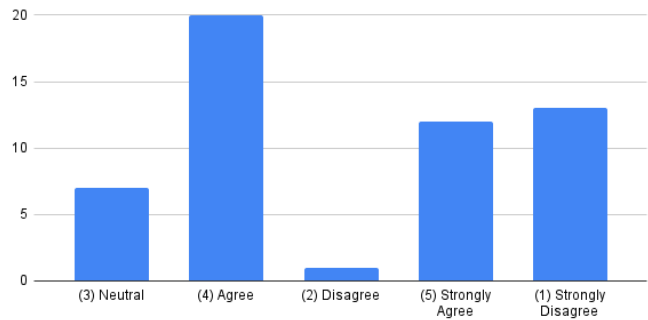


Figure 3.9 shows that dark mode positively influences overall user experience. Improved visual comfort and reduced eye strain appear to contribute to higher levels of user satisfaction.

Satisfaction with Dark Mode Compared to Light Mode

10. I believe dark mode is more satisfying to use compared to light mode.

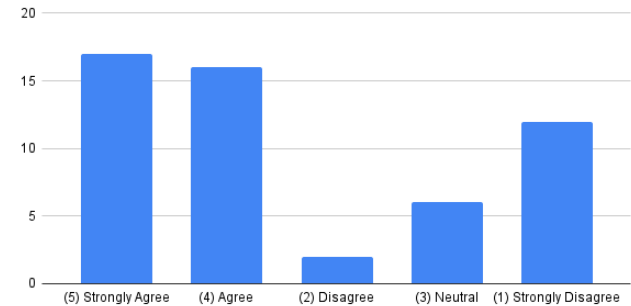


Figure 3.10 suggests that dark mode is generally perceived as more satisfying than light mode. This indicates that users associate dark mode with enhanced comfort, aesthetics, and usability.

CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of findings, conclusions, and recommendations of the study. It assesses the overall success of the research in relation to its stated objectives and goals. Furthermore, this chapter discusses the limitations encountered during the conduct of the study and provides directions for future research.

SUMMARY OF FINDINGS

The findings of the study reveal that the respondents are predominantly young individuals who are highly engaged in the use of digital technology and spend extended periods interacting with digital devices on a daily basis. Smartphones emerged as the primary devices used by respondents, indicating that most digital interactions occur on mobile platforms and in varying lighting environments. Due to prolonged screen exposure, respondents are more likely to experience eye strain and visual fatigue, which highlights the importance of display features that enhance visual comfort.

The results further show that a majority of respondents prefer using dark mode over light mode, particularly in low-light and nighttime conditions. Dark mode is perceived as effective in reducing eye strain, minimizing screen glare, and providing greater visual comfort during extended device usage. Respondents also associate dark mode with improved focus, relaxation of the eyes, and an overall more satisfying user experience. However, the findings also indicate that readability in dark mode may vary among users, suggesting that individual preferences, contrast levels, and interface design play a significant role in how display modes are perceived.

Overall, the study demonstrates that dark mode is generally favored by users and is perceived to positively influence comfort, usability, and satisfaction when using digital devices.

CONCLUSIONS

Based on the findings of the study, it is concluded that dark mode has a positive impact on users' visual comfort and overall experience when using digital devices. The preference for dark mode is largely attributed to its ability to reduce eye strain, lessen screen glare, and provide a more comfortable viewing experience, especially during prolonged use and in low-light environments.

The study also concludes that user preference for display mode is subjective and varies depending on individual comfort, visual sensitivity, and usage context. While dark mode offers several perceived benefits, light mode remains preferable for some users, particularly in terms of readability and familiarity. Therefore, providing flexibility and customization in interface design is essential to accommodate diverse user needs.

In general, the study confirms that dark mode serves as an effective interface option that enhances comfort, focus, and user satisfaction, as long as users are given the freedom to choose and adjust their preferred display settings.

RECOMMENDATIONS

In view of the conclusions drawn from the study, the following recommendations are presented:

For Application Developers

Application developers are encouraged to incorporate dark mode as an optional and customizable feature rather than a fixed default setting. Design considerations such as adjustable brightness, contrast, font size, and color balance should be implemented to improve readability and accommodate users with different visual preferences. Developers may also consider providing adaptive display modes that automatically adjust based on ambient lighting conditions to further enhance user comfort.

For Users

Users are advised to actively manage their display settings according to their environment, duration of use, and personal comfort. Switching between dark mode and light mode depending on lighting conditions may help reduce visual discomfort and eye strain. Users are also encouraged to practice healthy screen habits, such as taking regular breaks and adjusting screen brightness appropriately.

For Educational and Work Institutions

Educational institutions and workplaces may promote awareness of proper digital screen usage by encouraging the use of comfortable display settings and providing guidelines on reducing eye strain. Integrating ergonomic practices and digital wellness programs may help improve productivity and overall well-being among students and employees.

For Future Researchers

Future researchers may expand the scope of this study by involving a more diverse population in terms of age, occupation, and device usage. Further research may also

explore other interface design elements such as color combinations, font styles, and brightness levels, as well as conduct experimental studies to measure objective indicators of eye strain and visual fatigue.

REFERENCES

Buchner, A., & Baumgartner, N. (2007). Text–background polarity affects performance irrespective of ambient illumination and colour contrast.

<https://www.tandfonline.com/doi/abs/10.1080/00140130701306413>

Nielsen Norman Group. (2020). Dark mode: Benefits and drawbacks.

<https://www.nngroup.com/articles/dark-mode/>

Nielsen Norman Group. (2023). Dark mode: How users think about it and issues to avoid.

<https://www.nngroup.com/articles/dark-mode-users-issues>

Pathari, F. J. (2024). Dark vs. Light Mode on Smartphones: Effects on Eye Fatigue.

https://www.thinkmind.org/articles/achi_2024_3_150_20069.pdf

Android Developers. (n.d.). Dark theme.

<https://developer.android.com/develop/ui/views/theming/darktheme>

Sengsoon, P., & Intaruk, R. (2025). Immediate effects of light mode and dark mode features on visual fatigue in tablet users. International Journal of Environmental Research and Public Health. <https://www.mdpi.com/1660-4601/22/4/609>

Laine, J. (2025). Impact of Dark Mode on User Experience and Eye Comfort. Theseus Digital Repository. https://www.theseus.fi/bitstream/handle/10024/896088/Laine_Jere.pdf